

AKS Upgrade – Plan of Action Documentation

1. Pre-Requisites

- Verify supported Kubernetes versions.
- Review Kubernetes release notes.
- Validate deprecated APIs and breaking changes.
- Ensure lower environments are available for testing.
- Verify subnet IP availability.
- Validate cluster autoscaler configuration.

2. Verify AKS Versions

```
az aks get-versions --location eastus --output table  
az aks get-upgrades --resource-group aksdemo --name aksdemo --output table
```

3. Validate Breaking Changes Using Datree

```
curl https://get.datree.io | /bin/bash  
kubectl get all -A -o yaml > cluster-backup.yaml  
datree test cluster-backup.yaml --schema-version "1.30.0"
```

4. Upgrade Planning

```
DEV → QA/UAT → PROD  
Upgrade path:  
1.28 → 1.29 → 1.30
```

5. Networking and Capacity Validation

```
Minimum recommendation:  
• 5 free IP addresses  
az network vnet subnet show --resource-group <rg> --vnet-name <vnet> --name <subnet>
```

6. Configure Max Surge

```
az aks nodepool update --resource-group aksdemo --cluster-name aksdemo --name nodepool1 --max-surge 33%
```

7. Upgrade Lifecycle Events

- Surge – Create temporary node
- Drain – Evict pods from node
- Update – Upgrade node
- Delete – Remove temporary surge node

8. Upgrade Control Plane

```
az aks upgrade --resource-group aksdemo --name aksdemo --kubernetes-version 1.30.0 --control-plane-only
```

9. Upgrade Node Pools

```
az aks nodepool upgrade --resource-group aksdemo --cluster-name aksdemo --name nodepool1 --kubernetes-version 1.30.0
```

10. Post Upgrade Validation

```
kubectl get nodes
kubectl get pods -A
kubectl get events -A
kubectl top nodes
```

11. Rollback Strategy

AKS does not support direct downgrade.

Rollback approach:

- Restore backups
- Deploy workloads to new cluster
- Restore persistent data
- Validate applications

12. Production Checklist

Before Upgrade:

- Backup completed
- Maintenance window approved
- Monitoring enabled
- Stakeholders informed

After Upgrade:

- Validate applications
- Verify node versions
- Review cluster events